

Characterizing Galois extensions

Example: $M = \underbrace{\mathbb{Q}[x]}_{(x^3-2)} \ni \alpha = x + i(x^3-2)$

$$\text{Emb}_{\mathbb{Q}}(M, \mathbb{C}) \cong \{ \alpha \in \mathbb{C} : \alpha^3 = 2 \}$$

$$= \{ \sqrt[3]{2}, \zeta_3 \sqrt[3]{2}, \zeta_3^2 \sqrt[3]{2} \}$$

$$\begin{array}{ccc} \mathbb{Q}[x] & & \\ \downarrow & \searrow \text{ev}_{\sqrt[3]{2}} & \\ \tau_1: M & \longrightarrow & \mathbb{C} \\ \alpha & \longmapsto & \sqrt[3]{2} \end{array}$$

$$\parallel e^{2\pi i/3}$$

$$\tau_1(M) = \mathbb{Q}(\sqrt[3]{2}) \leq \mathbb{C}$$

$$\begin{array}{ccc} \tau_2: M & \longrightarrow & \mathbb{C} \\ \alpha & \longmapsto & \zeta_3 \sqrt[3]{2} \end{array}$$

$$\tau_2(M) = \mathbb{Q}(\zeta_3 \sqrt[3]{2}) \leq \mathbb{C}$$

$$\begin{array}{ccc} \tau_3: M & \longrightarrow & \mathbb{C} \\ \alpha & \longmapsto & \zeta_3^2 \sqrt[3]{2} \end{array}$$

$$\tau_3(M) = \mathbb{Q}(\zeta_3^2 \sqrt[3]{2}) \leq \mathbb{C}$$

these are distinct!

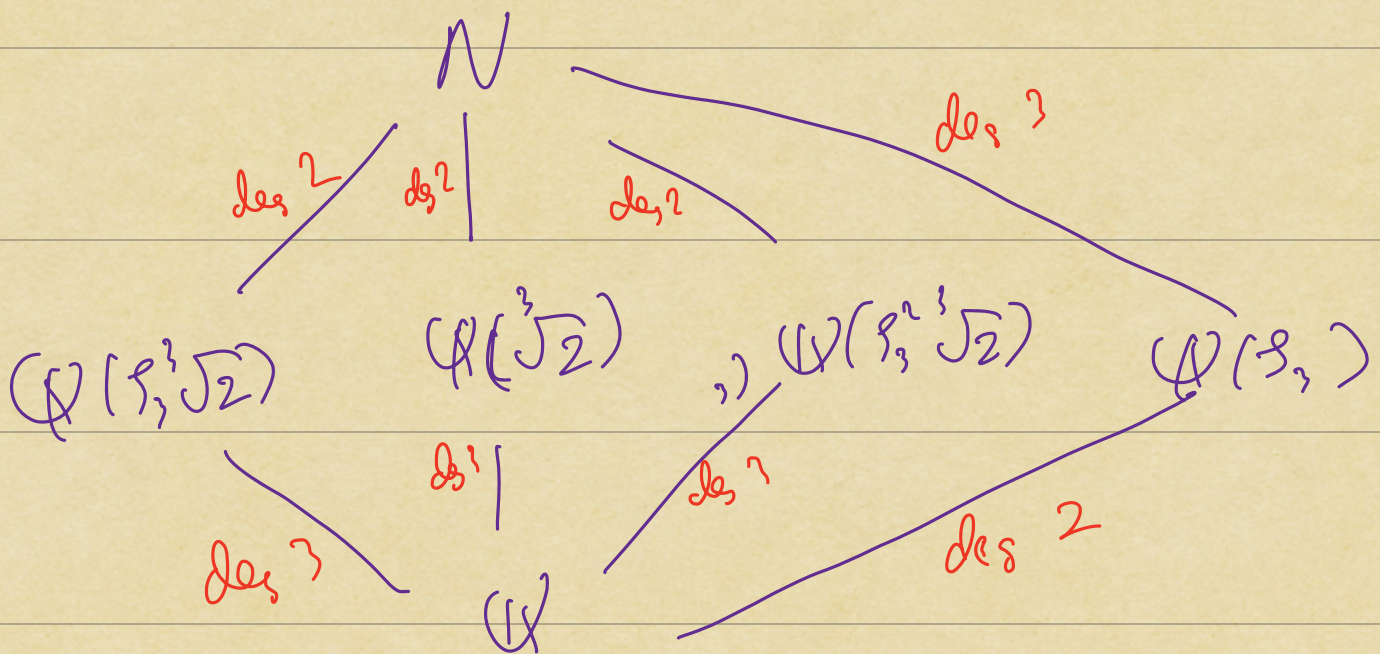
Consider

$$N = \mathbb{Q}(\sqrt[3]{2}, \zeta_3) \subseteq \mathbb{C}$$

$$\deg^2 \left\{ \begin{array}{l} (\mathbb{Q}(\sqrt[3]{2}))(\zeta_3) \\ \mathbb{Q}(\sqrt[3]{2}) \end{array} \right.$$

$$[N: \mathbb{Q}] = 6$$

$$\deg^3 \left\{ \begin{array}{l} 1 \\ \mathbb{Q} \end{array} \right.$$



$$\langle \sqrt[3]{2} \rangle \langle \rho_3 \rangle = N \longrightarrow N$$

$$\quad \quad \quad | \quad \quad \quad ||$$

$$M \simeq \langle \sqrt[3]{2} \rangle \xrightarrow[\text{or } \tau_3]{\tau_1 \text{ or } \tau_2} N$$

$$\quad \quad \quad |$$

✓ (1)

$$\text{Fix } i=1, 2, 3 \left[\begin{array}{l} i=1: \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ i=2: \sqrt[3]{2} \mapsto \rho_3 \sqrt[3]{2} \\ i=3: \sqrt[3]{2} \mapsto \rho_3^2 \sqrt[3]{2} \end{array} \right]$$

Pick $i=2$: $\sqrt[3]{2} \mapsto \rho_3 \sqrt[3]{2}$

$$\begin{array}{ccc} N & \cdots \xrightarrow{\quad} & N \\ | & \nearrow & | \\ M & \xrightarrow{\sigma} & \tau_2(M) \\ \alpha & \mapsto & \tau_2(\alpha) \end{array}$$

How many ways do I have to fill in this dotted arrow

Concretely: How many $\sigma: N \rightarrow N$ do we have s.t. $\forall \alpha \in M$,
 $\sigma(\alpha) = \tau_2(\alpha)$

$$N = M(\mathcal{F}_3) = \frac{M[x]}{(x^2 + x + 1)}$$

$$\text{Emb}_M(N, L) = \{ \beta \in L : \beta^2 + \beta + 1 = 0 \}$$

L/M : any
extension

$$\text{Emb}_M(N, N) \Rightarrow \{ \mathcal{F}_3, \mathcal{F}_3^2 \}$$

$\Rightarrow$ there are two $\sigma : N \rightarrow N$ with

$$\sigma(\alpha) = \tau_L(\alpha), \forall \alpha \in M$$

We can write down 6 elements

$$\text{in } \text{Emb}_{\mathbb{Q}}(N, N) = \text{Aut}(N/\mathbb{Q})$$

$$\left\{ \begin{array}{l} \sqrt[2]{2} \mapsto \sqrt[2]{2} \\ \mathcal{F}_3 \mapsto \mathcal{F}_3 \end{array} \right\}, \left\{ \begin{array}{l} \sqrt[2]{2} \mapsto \sqrt[2]{2} \\ \mathcal{F}_3 \mapsto \mathcal{F}_3^2 \end{array} \right\} = \tau$$

$$\sigma = \left\{ \begin{array}{l} \sqrt[2]{2} \mapsto \mathcal{F}_3 \sqrt[2]{2} \\ \mathcal{F}_3 \mapsto \mathcal{F}_3 \end{array} \right\}, \left\{ \begin{array}{l} \sqrt[2]{2} \mapsto \mathcal{F}_3 \sqrt[2]{2} \\ \mathcal{F}_3 \mapsto \mathcal{F}_3^2 \end{array} \right\}$$

$$\left\{ \begin{array}{l} \sqrt[2]{2} \mapsto \mathcal{F}_3^2 \sqrt[2]{2} \\ \mathcal{F}_3 \mapsto \mathcal{F}_3 \end{array} \right\}, \left\{ \begin{array}{l} \sqrt[2]{2} \mapsto \mathcal{F}_3^2 \sqrt[2]{2} \\ \mathcal{F}_3 \mapsto \mathcal{F}_3^2 \end{array} \right\}$$

$N/\mathbb{Q}$ is Galois,
and $|\text{Gal}(N/\mathbb{Q})| = 6$

Compute:

$$6 \circ \tau: \sqrt[3]{2} \xrightarrow{\tau} \sqrt[3]{2} \xrightarrow{6} \zeta_3 \sqrt[3]{2}$$

$$\parallel \quad \zeta_3 \xrightarrow{\tau} \zeta_3^2 \xrightarrow{6} \zeta_3^4$$

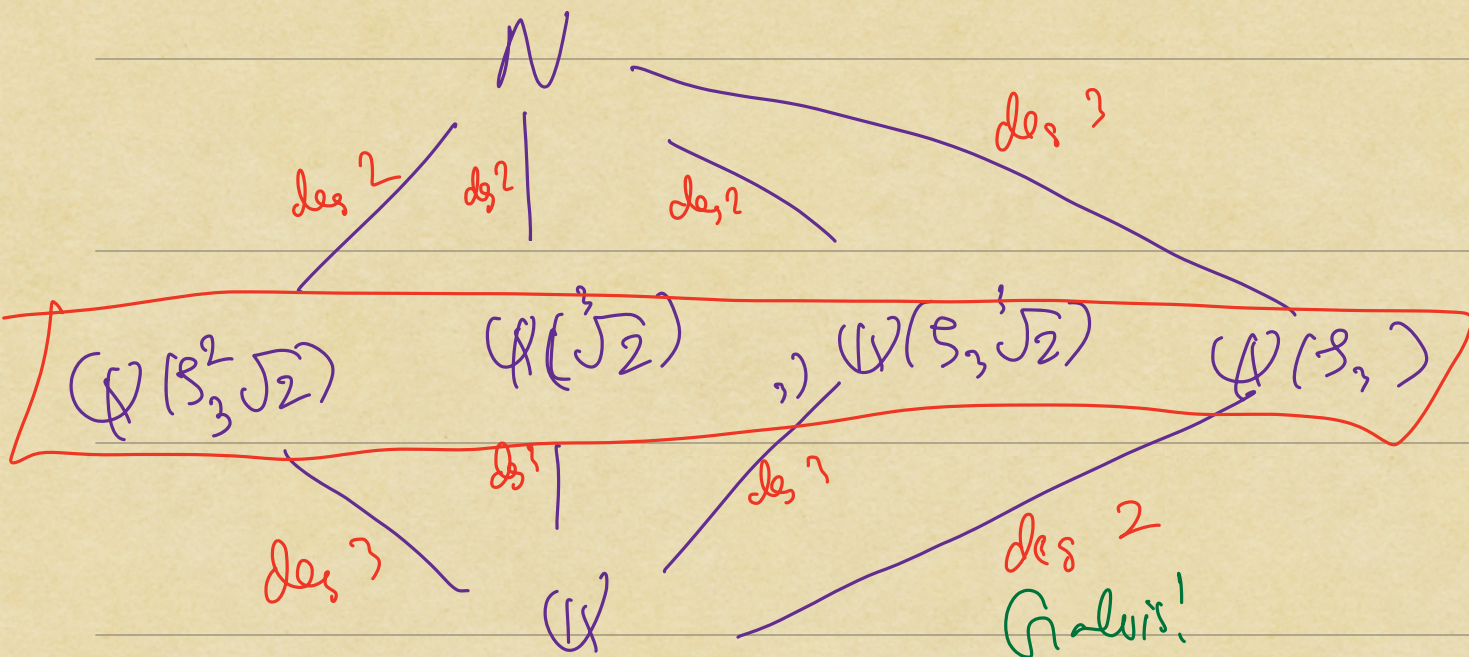
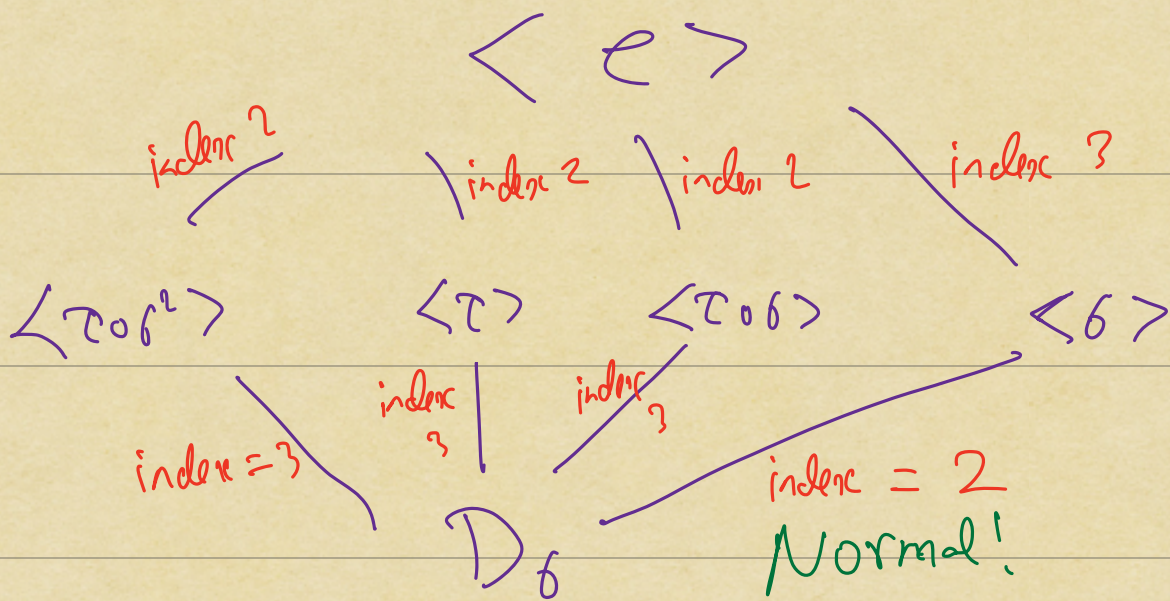
$$\tau \circ 6: \sqrt[3]{2} \xrightarrow{6} \zeta_3 \sqrt[3]{2} \xrightarrow{\tau} \zeta_3^2 \sqrt[3]{2}$$

$$\parallel$$

$$\tau(\zeta_3 \sqrt[3]{2}) = \tau(\zeta_3) \tau(\sqrt[3]{2})$$

$$\zeta_3 \xrightarrow{6} \zeta_3 \xrightarrow{\tau} \zeta_3^2$$

$$\Rightarrow \text{Gal}(N/\mathbb{Q}) \cong D_6$$



Thm M/K : finite extension

TFAE:

(1) M is Galois over K

(2) For all subextensions
 $M/L/K$

$$|\text{Emb}_K(L, M)| = [L:K]$$

(3) $\forall \alpha \in M$, with minimal poly,
 $m_\alpha(x) \in K[x]$ of deg d

$m_\alpha(x)$ has exactly d distinct
zeros in M

(4) $f(x) \in K[x]$ irred. of deg d

If $f(x)$ admits one zero in M

then $f(x)$ has d distinct zeros in M

(5) There exists a separable irreducible polynomial $f(x) \in K[x]$ of deg d s.t. (i) $f(x)$ has d distinct zeros in M $\alpha_1, \dots, \alpha_d \in M$
(ii) $M = K(\alpha_1, \dots, \alpha_d)$